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The Decimal System

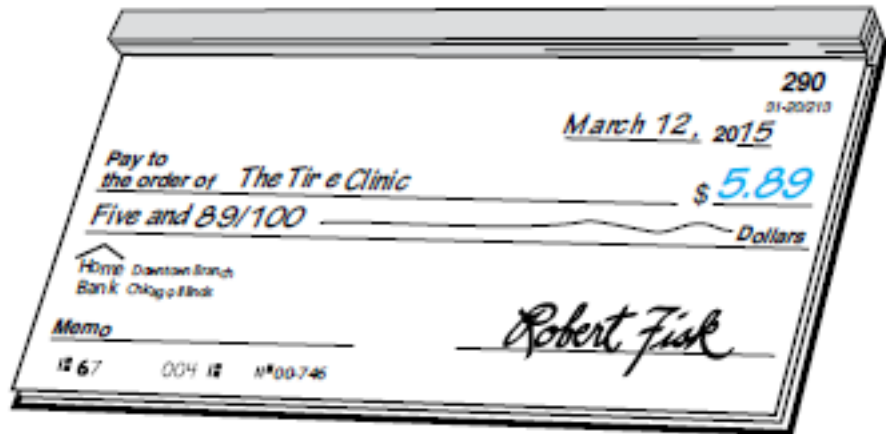
THE DECIMAL SYSTEM represents whole and decimal amounts. When a digit is written to the right of a decimal point, that digit has a value that is less than 1.

In the U.S. money system, parts of a dollar are shown using decimals. The amount \$5.75 means 5 whole dollars and 75 hundredths of a dollar.

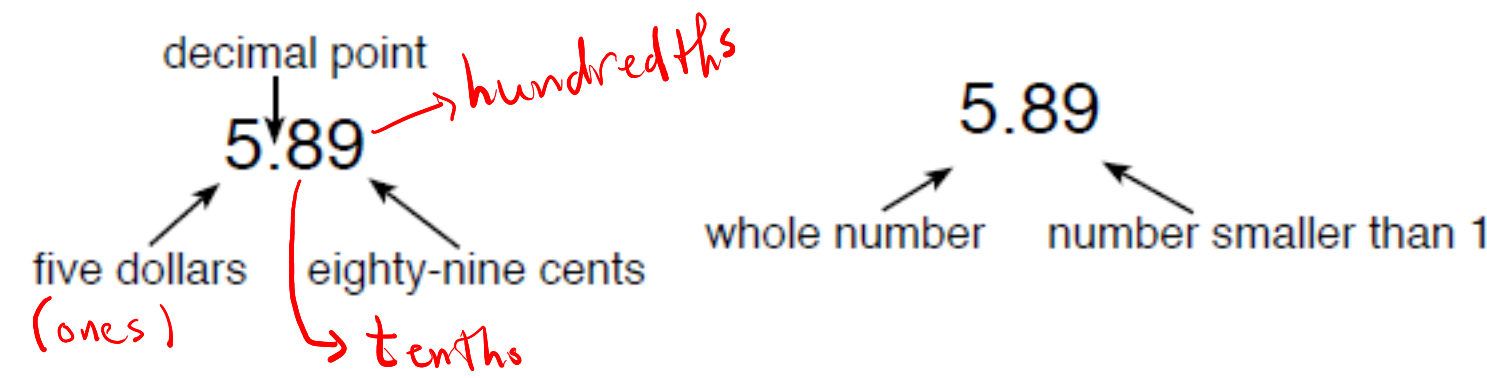
In this lesson, you'll apply decimals in many practical situations. You'll work with money and measurements. You'll see how the use of unit rates can help you save money when you shop. You will also see how estimating and using a calculator can help you become a better problem solver.

UNDERSTANDING DECIMALS

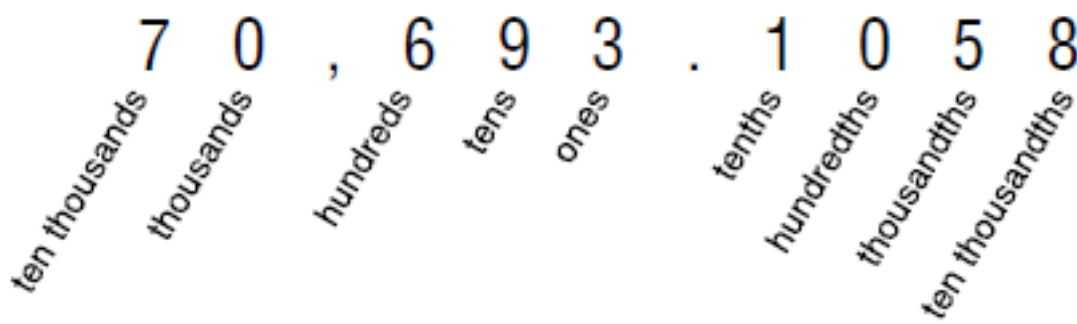
Each time you use money, you are working with decimals. In the U.S. money system, a **decimal point** separates dollars from cents. Numbers after the decimal point represent a value less than \$1.



Similarly, our number system uses the decimal point to separate whole numbers from numbers with a value *less than 1*.



A digit's position in relation to the decimal point (its place value) tells you how large or small it is. Look at the place value chart below.



What place is the 9 in? tens

What place is the 5 in? thousandths

It represents 9 tens, or 90.

It represents the fraction 5 thousandths, or $\frac{5}{1000}$.

There are two zeros in the number. Both are used as placeholders. In this number, there are no thousands and no hundredths.

Decimal values can be less than ten thousandths, just as whole numbers can be greater than ten thousands. The place value columns continue in both directions. Columns increase in value as they go to the left and decrease as they go to the right.

Example

[Hide](#)

Which is larger: 0.5 or 0.05?

Step 1: Look at the place value of the last digit in the decimal.

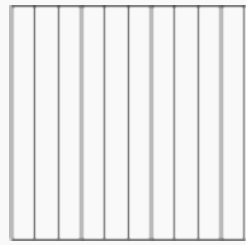
0.5

0.05

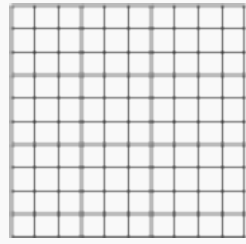
The place value is tenths.

The place value is hundredths.

Step 2: Think of a box divided into that many parts.

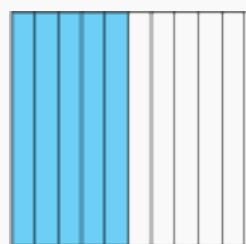


divided into tenths

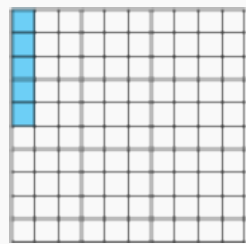


divided into hundredths

Step 3: Visualize the whole decimal as part of that box.



5 tenths (0.5)



5 hundredths (0.05)

5 tenths is larger than 5 hundredths.

0.5 > 0.05

Key Idea

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Numbers after the decimal point end in **ths**: *tenths*, *hundredths*, *thousandths*, and so on.

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Lesson 1: Number Sense

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